

Consider the curve along the unit circle,

$$Z(t) = e^{it} \quad (0 \leq t \leq 2\pi).$$

By definition of trigonometric,

$$\sin t = \frac{e^{it} - e^{-it}}{2i} = \frac{1}{2i} \left(Z - \frac{1}{Z} \right) \text{ and } \cos t = \frac{e^{it} + e^{-it}}{2} = \frac{1}{2} \left(Z + \frac{1}{Z} \right)$$

Using this facts, evaluate the integral,

$$\int_0^{2\pi} g(\sin \theta, \cos \theta) d\theta \text{ where } g \text{ is a rational function.}$$

Evaluate: for real numbers a, b with $|a| > |b| > 0$.

$$I = \int_0^{2\pi} \frac{1}{a + b \cdot \cos \theta} d\theta = \frac{2\pi}{\sqrt{a^2 - b^2}}.$$

Since $|a| > |b|$, $I = \frac{1}{b} \cdot \int_0^{2\pi} \frac{1}{\alpha/b + \cos \theta} d\theta = \frac{1}{b} \cdot \int_0^{2\pi} \frac{1}{\alpha + \cos \theta} d\theta$ where $\alpha = \frac{a}{b}$, $|\alpha| > 1$.

By the fact above, along C : $Z(\theta) = e^{i\theta}$ ($0 \leq \theta \leq 2\pi$),

$$\cos \theta = \frac{1}{2} \left(Z + \frac{1}{Z} \right) \text{ and } dz = i \cdot e^{i\theta} d\theta \Rightarrow d\theta = \frac{1}{ie^{i\theta}} dz = \frac{1}{i \cdot Z} dz.$$

$$\text{Therefore, } \int_0^{2\pi} \frac{1}{\alpha + \cos \theta} d\theta = \int_C \frac{1}{\alpha + \frac{1}{2} \left(Z + \frac{1}{Z} \right)} \cdot \frac{1}{iZ} dz = \frac{2}{i} \cdot \int_C \frac{1}{Z^2 + 2\alpha Z + 1} dz$$

Let $f(z) = \frac{1}{Z^2 + 2\alpha Z + 1}$. Since $Z^2 + 2\alpha Z + 1$ has two roots, $-\alpha \pm \sqrt{\alpha^2 - 1}$,

If $\alpha > 1$, then $-\alpha + \sqrt{\alpha^2 - 1}$ lies inside C , but $-\alpha - \sqrt{\alpha^2 - 1}$ lies outside of C .
 $(-\alpha + \sqrt{\alpha^2 - 1} < -\alpha + \sqrt{\alpha^2} = 0, \text{ but } (-1 < -\alpha + \sqrt{\alpha^2 - 1}).$

If $\alpha < -1$, similarly, $-\alpha + \sqrt{\alpha^2 - 1}$ lies outside C , $-\alpha - \sqrt{\alpha^2 - 1}$ lies inside of C .

So in case of $\alpha > 1$, $f(z)$ has only one isolated singularity $-\alpha + \sqrt{\alpha^2 - 1}$ as a simple pole. therefore the Residue Theorem gives that

$$\begin{aligned} \int_0^{2\pi} \frac{1}{\alpha + \cos \theta} d\theta &= \frac{2}{i} \cdot \int_C f(z) dz = \frac{2}{i} \cdot 2\pi i \cdot \lim_{z \rightarrow -\alpha + \sqrt{\alpha^2 - 1}} \left(Z - (-\alpha + \sqrt{\alpha^2 - 1}) \right) \cdot \frac{1}{Z^2 + 2\alpha Z + 1} \\ &= 4\pi \cdot \frac{1}{2\sqrt{\alpha^2 - 1}} = \frac{2\pi}{\sqrt{\alpha^2 - 1}}. \text{ Consequently} \end{aligned}$$

$$\int_0^{2\pi} \frac{1}{a + b \cos \theta} d\theta = \frac{1}{|b|} \cdot \frac{2\pi}{\sqrt{(\alpha/b)^2 - 1}} = \frac{2\pi}{\sqrt{a^2 - b^2}}.$$

Similarly, we can deduce that in case of $\alpha < -1$, as $= \frac{-2\pi}{\sqrt{a^2 - b^2}}$.

Subsequently, we can deduce that

$$I = \int_0^{\pi} \frac{1}{a^2 + \cos^2 \theta} d\theta = \int_0^{\pi} \frac{1}{a^2 + \sin^2 \theta} d\theta = \frac{\pi}{a\sqrt{1+a^2}} \quad \text{for } a > 0.$$

Using the useful formulas:

$$2 \cdot \cos^2 \theta = 1 + \cos 2\theta \quad \text{and} \quad 2 \cdot \sin^2 \theta = 1 - \cos 2\theta.$$

(Note: it can be deduced that $\cos(\alpha + \beta) = \cos \alpha \cdot \cos \beta - \sin \alpha \sin \beta$.)

$$\begin{aligned} \text{Now, } I &= \int_0^{\pi} \frac{1}{a^2 + \cos^2 \theta} d\theta = \int_0^{\pi} \frac{2}{1 + 2a^2 + \cos 2\theta} d\theta \quad \text{so, } 1 + 2a^2 > 1, \\ &= \int_0^{2\pi} \frac{1}{1 + 2a^2 + \cos u} du = \frac{2\pi}{\sqrt{(1 + 2a^2)^2 - 1}} = 2\pi \cdot \frac{1}{\sqrt{4a^4 + 4a^2}} \\ &= \frac{\pi}{a\sqrt{1+a^2}} \quad \square \end{aligned}$$

Evaluate

$$\int_0^{2\pi} \frac{1}{1 - 2\alpha \cdot \cos \theta + \alpha^2} d\theta = \frac{2\pi}{|\alpha^2 - 1|} \quad (\alpha \neq \pm 1, \text{ real}).$$

Proof. Notice that: for $z(\theta) = e^{i\theta}$, $\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} = \frac{1}{2} \cdot \left(z + \frac{1}{z}\right)$.

And, $dz = i \cdot e^{i\theta} d\theta \Rightarrow d\theta = -i \cdot z^{-1}$, so

$$\begin{aligned} \int_0^{2\pi} \frac{1}{1 - 2\alpha \cdot \cos \theta + \alpha^2} d\theta &= \int_C \frac{-i}{1 - \alpha \cdot \left(z + \frac{1}{z}\right) + \alpha^2} \cdot \frac{1}{z} dz \\ &= i \cdot \int_C \frac{1}{\alpha \cdot z^2 - (1 + \alpha^2) \cdot z + \alpha} dz \\ &= \frac{i}{\alpha} \cdot \int_C \frac{1}{z^2 - \left(\alpha + \frac{1}{\alpha}\right)z + 1} dz \end{aligned}$$

Since $z^2 - \left(\alpha + \frac{1}{\alpha}\right)z + 1 = (z - \alpha)\left(z - \frac{1}{\alpha}\right)$, consider two cases:

$-1 < \alpha < 1$ or $\alpha > 1$ or $\alpha < -1$, (if $\alpha = 0$, then it is just a constant integral).

If $-1 < \alpha < 1$, then the function in the integral has a simple pole at α

inside C , so $\text{Res}(f, \alpha) = \lim_{z \rightarrow \alpha} (z - \alpha) \cdot \frac{1}{(z - \alpha)\left(z - \frac{1}{\alpha}\right)} = \frac{1}{\alpha - \frac{1}{\alpha}} = \frac{\alpha}{\alpha^2 - 1}$.

Now, by the Residue theorem,

$$\int_0^{2\pi} \frac{1}{1 - 2\alpha \cos \theta + \alpha^2} d\theta = \frac{i}{\alpha} \cdot \int_C \frac{1}{z^2 - \left(\alpha + \frac{1}{\alpha}\right)z + 1} dz = \frac{i}{\alpha} \cdot 2\pi i \cdot \frac{\alpha}{\alpha^2 - 1} = \frac{2\pi}{1 - \alpha^2}.$$

If $\alpha > 1$, then the function has a simple pole at $\frac{1}{\alpha}$, inside C , so

$$\text{Res}\left(f, \frac{1}{\alpha}\right) = \lim_{z \rightarrow \frac{1}{\alpha}} \left(z - \frac{1}{\alpha}\right) \cdot \frac{1}{(z - \alpha)\left(z - \frac{1}{\alpha}\right)} = \frac{1}{\frac{1}{\alpha} - \alpha} = \frac{\alpha}{1 - \alpha^2}.$$

therefore, by the Residue theorem again,

$$= \frac{2\pi}{\alpha^2 - 1}.$$

If $\alpha < -1$, we obtain the value, similarly $\alpha > 1$ case.

Consequently,
$$\int_0^{2\pi} \frac{1}{1 - 2\alpha \cdot \cos \theta + \alpha^2} d\theta = \frac{2\pi}{|\alpha^2 - 1|} \quad \square$$

Evaluate

$$\int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin^2 \theta} d\theta = \frac{\pi}{2\sqrt{2}} .$$

Proof.